

LAB 9 KNN CLASSIFICATION

```
> install.packages("dplyr")
> install.packages("caTools")
> install.packages("class")
> install.packages("caret")
> library(dplyr)
> library(caTools)
> library(class)
> library(caret)

> data("iris")
> summary(iris)

   Sepal.Length   Sepal.Width   Petal.Length   Petal.Width
Min.   :4.300   Min.    :2.000   Min.    :1.000   Min.    :0.100
1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300
Median :5.800   Median :3.000   Median :4.350   Median :1.300
Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
Max.   :7.900   Max.    :4.400   Max.    :6.900   Max.    :2.500

   Species
setosa   :50
versicolor:50
virginica :50

> str(iris)
'data.frame': 150 obs. of 5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species      : Factor w/ 3 levels "setosa","versicolor",...: 1 1 1 1 1 1 1 1 1 1 ...

> split<-sample.split(iris,SplitRatio = 0.7)
> train<-subset(iris,split==TRUE)
> test<-subset(iris,split==FALSE)
```

```
> train_scale<-scale(train[,1:4])
> test_scale<-scale(test[,1:4])
```

```
> classifier_knn =knn(train=train_scale,test=test_scale,k=3,cl=train$Species)
```

```
> classifier_knn
```

```
[1] setosa      setosa      setosa      setosa      setosa      setosa
[7] setosa      setosa      setosa      setosa      setosa      setosa
[13] setosa      setosa      setosa      setosa      setosa      setosa
[19] setosa      setosa      versicolor versicolor versicolor versicolor
[25] versicolor versicolor versicolor versicolor virginica  virginica
[31] versicolor virginica  versicolor versicolor versicolor versicolor
[37] versicolor versicolor versicolor versicolor virginica  virginica
[43] virginica  virginica  virginica  virginica  virginica  virginica
[49] virginica  virginica  virginica  virginica  virginica  virginica
[55] virginica  virginica  virginica  virginica  virginica  virginica
Levels: setosa versicolor virginica
```

```
> cm<-table(test$Species,classifier_knn)
```

```
> cm
```

	classifier_knn		
	setosa	versicolor	virginica
setosa	20	0	0
versicolor	0	17	3
virginica	0	0	20

```
> report<-confusionMatrix(classifier_knn,test$Species)
```

```
> report
```

Confusion Matrix and Statistics

	Reference		
Prediction	setosa	versicolor	virginica
setosa	20	0	0
versicolor	0	17	0
virginica	0	3	20

Overall Statistics

Accuracy : 0.95

95% CI : (0.8608, 0.9896)

No Information Rate : 0.3333

P-Value [Acc > NIR] : < 2.2e-16

Kappa : 0.925

McNemar's Test P-Value : NA

Statistics by Class:

	Class: setosa	Class: versicolor	Class: virginica
Sensitivity	1.0000	0.8500	1.0000
Specificity	1.0000	1.0000	0.9250
Pos Pred Value	1.0000	1.0000	0.8696
Neg Pred Value	1.0000	0.9302	1.0000
Prevalence	0.3333	0.3333	0.3333
Detection Rate	0.3333	0.2833	0.3333
Detection Prevalence	0.3333	0.2833	0.3833
Balanced Accuracy	1.0000	0.9250	0.9625

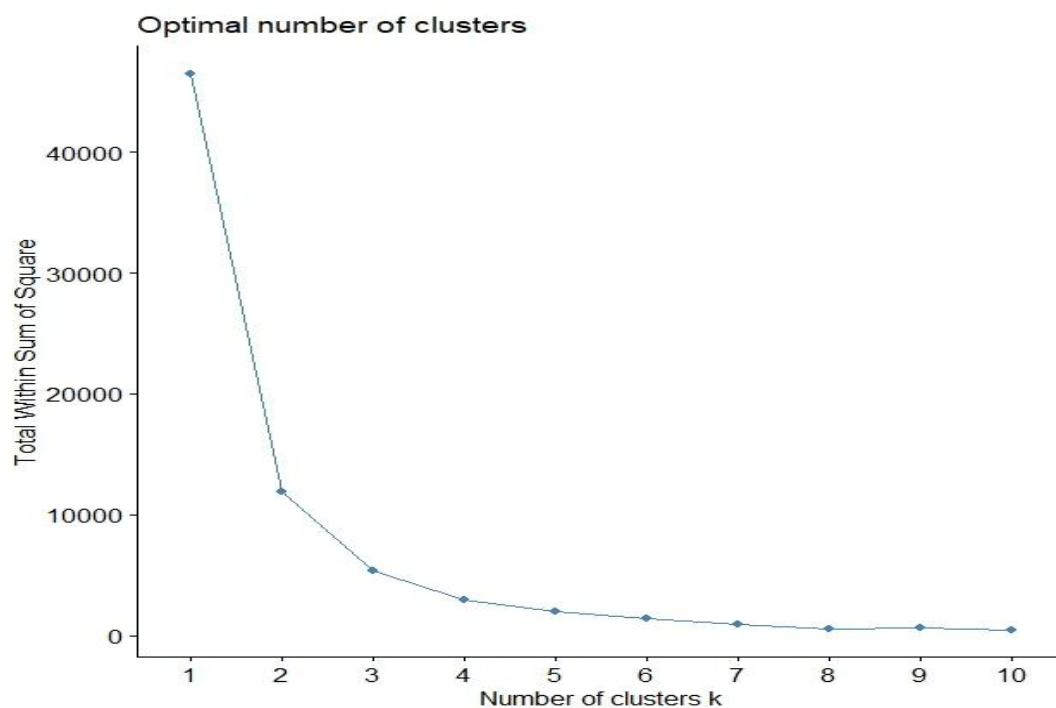
K MEANS CLUSTERING

```
> install.packages("factoextra")
> install.packages("cluster")
> install.packages("dplyr")
> library(dplyr)
> library(factoextra)
> library(cluster)

> f=read.csv(file.choose())

> data=f[,c("weight", "size", "color_score")]
```

```
> fviz_nbclust(data,kmeans,method ="wss")
```



```
> km<-kmeans(data,centers = 3)
```

```
> fviz_cluster(km, data = data, main = "K-means cluster")
```

